**Candidate's Name: ……..……..**

**Signature: ……..**

| STREAM: | | | | | | | |
|---|---|---|---|---|---|---|---|

**525/1**
**CHEMISTRY**
**March 2026**
**$2\frac{1}{2}$ hours**

# PRE-REGISTRATION ASSESSMENT 2026

*Uganda Advanced Certificate of Education*

**S.6 CHEMISTRY**

*Paper 1*

Set I

**2 hours 30 minutes**

**INSTRUCTIONS TO CANDIDATES**:

*This paper has* **two** *sections* **A** *and* **B.**

*Section* **A** *has two compulsory items while* **B** *has two parts* **Part I** *and* **Part II**

*Each of part I and part II has two items, Answer only* **one** *from each.*

*Answers to Section* **A** ***must*** *be written in the spaces provided and Section* **B** ***must*** *be written in the answer booklet(s) provided*

*Answer four in all.*

*Where necessary use,*

*Molar gas volume at s.t.p = 22.4dm$^3$*

| FOR EXAMINER'S USE ONLY | | |
|---|---|---|
| **ITEM** | **CODE** | **SCORE** |
| **1** | | |
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| **TOTAL** | | |

# SECTION A

*Answer* **all** *questions in this section*

## ITEM 1

PhosGuard Industries manufacture phosphorus-derived reagents used in water treatment to remove heavy metal poisoning. Part of their continuous firing reactor line handles phosphorus(V) chloride ($PCl_5$) as an intermediate. The safety team performs gas phase equilibrium checks to ensure plant safety and downstream water treatment compatibility.

A gas phase equilibrium test was carried out by introducing *3.60 moles* of phosphorus(V) chloride into a rigid vessel of volume *2.00 dm³* at a given temperature. Analysis showed 1% dissociation according to:

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g) \quad \Delta H = +124\ kJ\ mol^{-1}.$$

Using this information, the equilibrium constant Kc can be determined.

In a follow-up process optimization test, the company introduces an additional 0.050 moles of chlorine gas ($Cl_2$) into the same system at the same temperature.

A treatment test was carried out where *0.208 g* of phosphorus(V) chloride ($PCl_5$) was completely hydrolysed in *1.00 dm³* of water, producing hydrochloric acid (HCl).

The acid produced supplies chloride ions ($Cl^-$), which are then used to react with lead(II) ions ($Pb^{2+}$) in wastewater to form a precipitate, equations involved;

$$PCl_5\ (g) + 4H_2O\ (l) \rightarrow H_3PO_4\ (aq) + 5HCl(aq)$$

$$Pb^{2+}(aq) + 2Cl^-(aq) \rightarrow PbCl_2(s)$$

A sample of wastewater contains 0.020 mol $dm^{-3}$ $Pb^{2+}$ ions. You have been contacted for help.

Task:

As a learner of chemistry, guide the company on how to:

(a) Calculate the equilibrium constant Kc and comment on the extent of the reaction.

| | $PCl_5(g)$ | $\rightleftharpoons$ | $PCl_3(g)$ | + | $Cl_2(g)$ |
|---|---|---|---|---|---|
| ***Initial moles*** | $\mathbf{3.6}$ | | $\mathbf{0}$ | | $\mathbf{0}$ |
| *Moles dissociated/formed* | $3.6\ \alpha$ | | $3.6\ \alpha$ | | $3.6\ \alpha$ |
| ***Moles at equilibrium*** | $\mathbf{3.6(1-\alpha)}$ | | $\mathbf{3.6\alpha}$ | | $\mathbf{3.6\alpha}$ |
| ***Concentration at equilibrium*** | $\frac{\mathbf{3.6(1-\alpha)}}{\mathbf{2.0}}$ | | $\frac{\mathbf{3.6\alpha}}{\mathbf{2.0}}$ | | $\frac{\mathbf{3.6\alpha}}{\mathbf{2.0}}$ |

$$But\ \alpha = {}^{1}/_{100} = \mathbf{0.01}$$

$$\therefore [PCl_5] = \frac{3.6(1-0.01)}{2.0} = \mathbf{1.782M}\ ;\quad [PCl_3] = \frac{3.6\times 0.01}{2.0} = \mathbf{0.018M};\quad [Cl_2] = \frac{3.6\times 0.01}{2.0} = \mathbf{0.018M}$$

$$\boldsymbol{K_C = \frac{[PCl_3][Cl_2]}{[PCl_5]}}\quad \frac{\boldsymbol{moldm^{-3}\times moldm^{-3}}}{\boldsymbol{moldm^{-3}}}$$

$$\boldsymbol{K_C = \frac{(0.018)^2}{1.782} = 1.82\times 10^{-4}\ moldm^{-3}}$$

*∴The value of Kc is* ***very small,*** *hence equilibrium lies to the* ***left*** *and the reaction proceeds to a* ***small extent.***

(b) Determine the new equilibrium concentrations of all species during optimization test, hence determine the direction in which the equilibrium shifts and explain your answer.

$Moles\ of\ PCl_5 = 3.6(1-0.01) = \mathbf{3.564}$; $Moles\ of\ PCl_3 = 3.6 \times 0.01 = \mathbf{0.036}$

$Moles\ of\ Cl_2 = 3.6 \times 0.01 = \mathbf{0.036}$ ; *On addition of* 0.050 *moles of* $Cl_2$ *in the reaction vessel*

$\therefore$ ***new*** $\mathbf{Cl_2 = 0.036 + 0.050 = 0.086}$ ***moles***

***Equilibrium shifts to the left to reduce excess*** $\mathbf{Cl_2}$; *Let* $\boldsymbol{x}$ *be the amount reacting*:

| | $PCl_3(g)$ | + $Cl_2(g)$ | ⇌ $PCl_5(g)$ |
|---|---|---|---|
| ***New Initial moles*** | $\mathbf{0.036}$ | $\mathbf{0.086}$ | $\mathbf{3.564}$ |
| *Moles reacted/formed* | $\boldsymbol{x}$ | $\boldsymbol{x}$ | $\boldsymbol{x}$ |
| ***Moles at equilibrium*** | $\mathbf{(0.036 - x)}$ | $\mathbf{(0.086 - x)}$ | $\mathbf{(3.564 + x)}$ |
| ***Concentration at equilibrium*** | $\frac{\mathbf{(0.036 - x)}}{\mathbf{2.0}}$ | $\frac{\mathbf{(0.086 - x)}}{\mathbf{2.0}}$ | $\frac{\mathbf{(3.564 + x)}}{\mathbf{2.0}}$ |

*But Backward New* $K_{C_{New}} = \frac{1}{K_C}$

$= \frac{1}{1.82 \times 10^{-4}} \frac{1}{moldm^{-3}}$

$= \mathbf{5494.51\ mol^{-1}dm^3}$

$K_C = \frac{[\mathbf{PCl_5}]}{[\mathbf{PCl_3}][\mathbf{Cl_2}]}$

$5494.51 = \frac{\frac{(3.564+x)}{2.0}}{\left(\frac{(0.036-x)}{2.0}\right)\left(\frac{(0.086-x)}{2.0}\right)}$

$5494.51[(0.036 - x)(0.086 - x)] = 2(3.564 + x)$

$5494.51(0.003096 + x^2 - 0.122x) = 2(3.564 + x)$

$17.01 + 5494.51x^2 - 670.33x = 7.128 + 2x$

$5494.51x^2 - 672.33x + 9.882 = 0$

$x = \frac{672.33 \pm \sqrt{(672.33)^2 - 4(5494.51)(9.882)}}{2(5494.51)}$

$x = 0.105\ or\ 0.0171$

$x$ *must be less than* $\mathbf{0.036}$ *and* $\mathbf{0.086}$ *(cannot exceed initial moles),*

$\therefore\ \boldsymbol{x} = \mathbf{0.0171}$ ***moles***

***At new equilibrium***;

$[PCl_5] = \frac{(3.564 + 0.0171)}{2.0} = \mathbf{1.791 moldm^{-3}}$;

$[PCl_3] = \frac{(0.036-0.0171)}{2.0} = \mathbf{0.00945 moldm^{-3}}$;

$[\mathbf{Cl_2}] = \frac{(0.086-x)}{2.0} = \mathbf{0.0345\ moldm^{-3}}$

$\therefore$ ***Equilibrium shifts to the left to reduce excess*** $\mathbf{Cl_2}$ ***introduced at equilibrium***

(c) Explain what would happen to the equilibrium position, the equilibrium constant (Kc) and the speed of attainment of equilibrium when the following changes are made:

i. Increasing the temperature

*According to Le Chatelier's principle, the forward reaction is* ***endothermic (ΔH = +124 kJ/mol)****. Increasing temperature adds heat to the system, therefore the equilibrium shifts* ***from left to right*** *to absorb the added heat, producing more* $PCl_3$ *and* $Cl_2$*. The value of* ***Kc increases*** *because equilibrium constant depends on temperature and increases for endothermic reactions. The* ***rate of attainment of equilibrium increases*** *since higher temperature increases kinetic energy and collision frequency of molecules.*

ii. Decreasing the pressure

*According to Le Chatelier's principle, the reaction involves an* ***increase in number of gas molecules (1 mole → 2 moles)****. Decreasing pressure shifts equilibrium* ***from left to right****, favouring the side with more gas molecules to increase pressure. The value of* ***Kc remains unchanged*** *because pressure does not affect equilibrium constant. The* ***rate of attainment of equilibrium decreases*** *since fewer particles per unit volume result in fewer effective collisions.*

iii. Adding helium gas at constant volume

*According to´Le Chatelier's principle, helium is an* ***inert gas*** *and does not take part in the reaction. Adding helium at constant volume does* ***not change the concentration or partial pressure of reacting gases****, therefore the* ***equilibrium position remains unchanged****. The value of* ***KcK_cKc remains unchanged****, and the* ***rate of attainment of equilibrium remains unchanged*** *since there is no effect on reacting particles.*

(d) Advise the company on how to optimize operating conditions (temperature, pressure, concentration) in order to, control equilibrium effectively, improve production efficiency and ensure safety in handling phosphorus(V) chloride

- *High temperature increases yield of products but excessive heat may be dangerous, therefore moderate temperature should be used.*
- *Low pressure favors forward reaction but must be controlled to avoid gas leakage.*
- *Continuous removal of $Cl_2$ shifts equilibrium to the right increasing production.*
- *Proper monitoring and closed systems prevent leakage of toxic gases like chlorine and phosphorus(V) chloride.*

(e) Calculate the maximum number of moles of $Pb^{2+}$ ions that can be precipitated by the chloride ions present, Comment whether chloride ions are sufficient to completely remove $Pb^{2+}$ ions from the wastewater.

$$Molar\ mass\ of\ PCl_5 = 208.5g$$

$$0.208\ g\ contain\ \frac{0.208}{208.5} = 0.0010\ moles\ of\ PCl_5$$

***From equation***: $\boldsymbol{PCl_5 + 4H_2O \rightarrow H_3PO_4 + 5HCl}$

$$1\ mole\ of\ PCl_5\ produces\ 5\ moles\ of\ Cl^-$$

$$0.0010\ moles\ of\ PCl_5\ produce\ \frac{(0.0010 \times 5)}{1} = \boldsymbol{0.0050\ moles\ of\ Cl^-}$$

***From equation:*** $\boldsymbol{Pb^{2+}(aq) + 2Cl^-(aq) \rightarrow PbCl_2(s)}$

$$2\ moles\ of\ Cl^-\ react\ with\ 1\ mole\ of\ Pb^{2+}$$

$$0.0050\ moles\ of\ Cl^-\ react\ with\ \frac{0.0050}{2} = \boldsymbol{0.0025\ moles\ of\ Pb^{2+}}$$

$$1000\ cm^3\ a\ sample\ of\ wastewater\ contains\ \boldsymbol{0.020\ moles\ of\ Pb^{2+}}$$

$$\therefore\ \boldsymbol{chloride\ ions\ remove\ only\ 0.0025\ moles}$$

$$\therefore\ \boldsymbol{chloride\ ions\ are\ not\ sufficient\ to\ completely\ remove\ Pb^{2+}}$$

(f) Suggest possible environmental impacts PhosGuard Industries should beware of and propose appropriate mitigation measures.

*Leakage of chlorine gas during processing can cause* ***respiratory problems*** *and* ***air pollution*** *since it is highly* ***toxic,*** *hence hazardous to human* ***health****. This can be mitigated by* ***installing gas scrubbers*** *and ensuring closed systems.*

*Discharge of* ***acidic solutions*** *into water bodies can* ***lower pH*** *and* ***harm aquatic life****. This is controlled by* ***neutralizing the effluents*** *before discharge.*

*Improper disposal of lead compounds can cause environmental contamination specifically* ***soil pollution reduced fertility, plant yield associated food poisoning****. This can be mitigated by proper collection, recycling, and safe disposal of sludge.*

## Item 2

A chemical processing plant plans to scale up production of propan-2-ol, an important solvent used in sanitizers. Plant managers are evaluating two possible preparation routes, and the management wants a full thermochemical assessment before commissioning either pathway. Major preparation involves hydrogenation of Propanone:

$$CH_3\overset{O}{\overset{\|}{C}}CH_3 + H_2(g) \longrightarrow CH_3\overset{OH}{\overset{|}{C}}HCH_3(l)$$

Alternatively, propan-2-ol can be obtained from hydration of propene:

$$CH_3CH = CH_2(g) + H_2O(l) \longrightarrow CH_3\overset{OH}{\overset{|}{C}}HCH_3(l)$$

The thermochemical department has supplied the following bond energies and auxiliary standard enthalpy information for use:

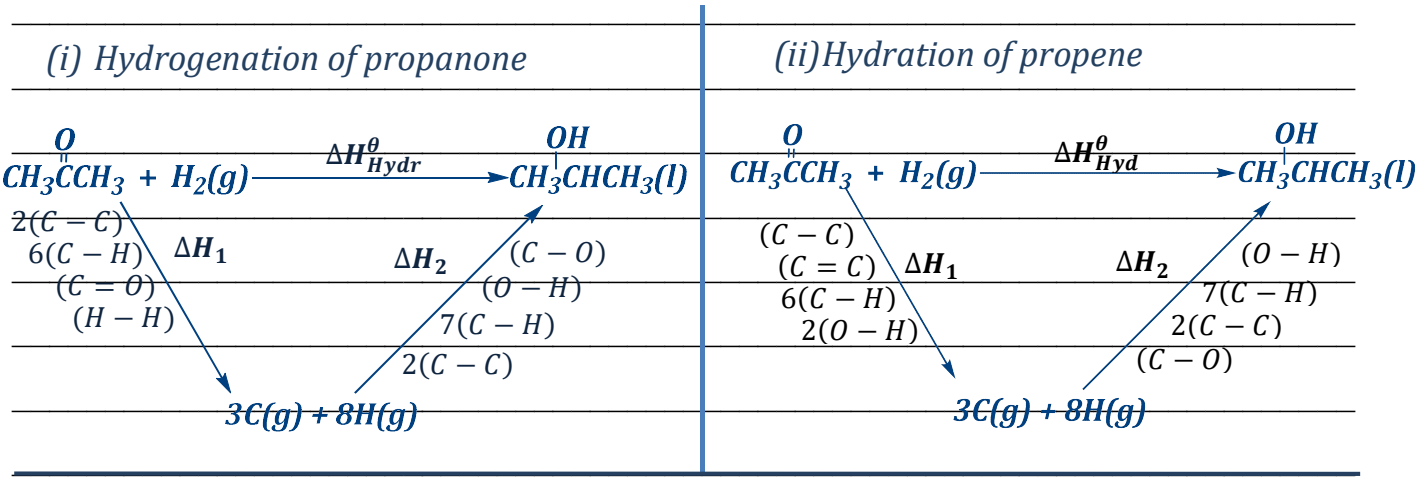
| Bond / Data | C=O | C=C | C–C | C–H | C–O | O–H | H–H |
|---|---|---|---|---|---|---|---|
| Value (kJ·mol$^{-1}$) | 715.41 | 614.28 | 345.01 | 414.50 | 351.11 | 464.23 | 436.43 |

To inform decision making for the best preparation reaction the manager has been advised by the expert that the best reaction must be feasible therefore you have been contacted for help.

**Task:**

As a learner of chemistry, help the manager

(a) For each route, construct a Hess's law energy cycle, using the thermochemical data.




(b) Use the energy cycles above to calculate the enthalpy change for:

(i) Hydrogenation of propanone

$$\Delta H^{\theta}_{Hydr} = \Sigma \text{ bond energies broken} - \Sigma \text{ bond energies made}$$

$$\Delta H^{\theta}_{Hydr} = \Delta H_1 - \Delta H_2$$

$$\Delta H^{\theta}_{Hydr} = [2(C-C) + 6(C-H) + (C=O) + (H-H)] - [2(C-C) + 7(C-H) + (O-H) + (C-O)]$$

$$\Delta H^{\theta}_{Hydr} = [2 \times 345.01 + 6 \times 414.50 + 715.41 + 436.43] - [2 \times 345.01 + 7 \times 414.50 + 464.23 + 351.11]$$

$$= 4328.86 - 4406.86$$

$$\therefore \Delta H^{\theta}_{Hydr} = -78\ kJmol^{-1};\ \therefore \textbf{Reaction is exothermic}$$



***Turn Over***

(ii) Hydration of propene

$$\Delta H^{\theta}_{Hydr} = \Sigma\ bond\ energies\ broken - \Sigma\ bond\ energies\ made$$

$$\boldsymbol{\Delta H^{\theta}_{Hydr} = \Delta H_1 - \Delta H_2}$$

$$\Delta H^{\theta}_{Hydr} = [(C-C) + (C=C) + 6(C-H) + 2(O-H)] - [2(C-C) + 7(C-H) + (O-H) + (C-O)]$$

$$\Delta H^{\theta}_{Hydr} = [345.01 + 614.28 + 6 \times 414.50 + 2 \times 464.23] - [2 \times 345.01 + 7 \times 414.50 + 464.23 + 351.11]$$

$$= 4374.75 - 4406.86$$

$$\therefore \boldsymbol{\Delta H^{\theta}_{Hydr} = -32.11\ kJmol^{-1};\ \therefore\ Reaction\ is\ exothermic}$$

(c) Evaluate which preparation route is more feasible for large scale production of propan-2-ol in the factory.

*According to thermochemical principles, a more feasible reaction is one that is* ***more exothermic (more negative ΔH)*** *since it releases more energy and is energetically favorable. Hydrogenation:* $\boldsymbol{\Delta H = -78.0\ kJ\ mol^{-1}}$ *while Hydration:* $\boldsymbol{\Delta H = -32.11\ kJ\ mol^{-1}}$ *.:Hydrogenation of propanone is more feasible; This is because it releases* ***more heat energy****, making it more favorable for large-scale production.*

(d) Predict any major environmental impact arising from the process and propose a realistic mitigation measure for the impact.

*Hydrogenation involves the use of hydrogen gas which is* ***highly flammable****. Leakage during processing can cause explosions and fire outbreaks, posing danger to workers and the environment. This can be mitigated by* ***using airtight reactors and regular monitoring of pipelines****.*

*Industrial processes may release* ***volatile organic compounds (VOCs)*** *which contribute to air pollution. This can be controlled by installing* ***gas capture and treatment systems*** *to reduce emissions.*

## SECTION B

### Part I

*Attempt* **One** *item in this section*

**Item 3**

Your chemistry class has been invited by a chemical plant management near Lake Victoria to assist in identifying ways of optimizing the production of key industrial compounds including sodium oxide, magnesium oxide and silicon dioxide used in glassmaking and ceramics. The plant management has observed inconsistencies in melting points and reactivity, which are affecting product quality and safety. You are tasked with conducting a scientific investigation to analyze periodic trends of the elements, compound properties, and molecular structures to recommend improvements.

Important chemical data of the findings about the elements and their compounds is provided in the table below to assist in the analysis.

| **Element** | **Atomic Number** | **Atomic Radius (pm)** | **Ionisation Energy (kJ/mol)** | **Melting Point (°C)** | **Oxide Melting Point (°C)** |
|---|---|---|---|---|---|
| Sodium (Na) | 11 | 186 | 496 | 98 | 1275 |
| Magnesium (Mg) | 12 | 160 | 738 | 650 | 2800 |
| Aluminium (Al) | 13 | 143 | 578 | 660 | 2072 |
| Silicon (Si) | 14 | 118 | 786 | 1410 | 1710 |
| Phosphorus (P) | 15 | 110 | 1012 | 44 | 580 |
| Sulphur (S) | 16 | 104 | 1000 | 115 | Gas/Sublimes |
| Chlorine (Cl) | 17 | 99 | 1251 | -101 | Gas/Gas |

**Task**

As a chemistry learner, write a report about the periodic trends of the elements and their oxides, and recommend improvements to the plant's production processes.

**Item 4**

Uganda faces persistent challenges in rural electrification. Many communities rely on unreliable and costly energy sources like diesel generators, kerosene lamps, and car batteries. A Ugandan company is developing solar-powered micro-grids and has invited your chemistry class to help identify suitable materials for solar batteries and wiring. These materials must be affordable, corrosion-resistant, and durable under Uganda's rural conditions.

The company is investigating elements similar to those in Periods 2 and 3 of the Periodic Table. The measured properties of selected elements are shown below:

| Element | Atomic Radius (pm) | First Ionization Energy (kJ/mol) | Electronegativity (Pauling) | Typical Bonding Type |
|---|---|---|---|---|
| Li | 145 | 520 | 1.0 | Metallic |
| Mg | 130 | 730 | 1.5 | Metallic |
| Al | 110 | 1000 | 2.5 | Metallic |
| Si | 95 | 1250 | 3.0 | Covalent |
| P | 85 | 1500 | 3.5 | Covalent |

**Task**

**As a chemistry learner;**

Use your understanding of atomic structure, periodic trends, chemical bonding and the information provided to propose suitable materials for solar batteries and wiring in Uganda's rural micro-grid systems

## Part II

*Attempt* **One** *item in this section*

**ITEM 5**

A certain town in central Uganda faces growing concerns over water pollution from agro-processing industries, especially during the cocoa harvest season. Wastewater from cocoa processing plants often contains organic matter including acids, and amines that can harm aquatic life if released untreated.

A group of entrepreneurs is setting up a small plant to process cocoa husks, a by-product of cocoa bean production, into value-added products. Chemical analysis of the husks shows that they contain ethanol (from natural fermentation), ethanoic acid, small amounts of amines (from protein breakdown), and aromatic aldehydes such as vanillin(4-hydroxy-3-methoxybenzaldehyde).

To address the water pollution challenge and create additional revenue streams, the team plans to:

1. Determine the relationship between the structure of vanillin and ethanoic acid, and their solubility.
2. Convert ethanol into amines for use in water treatment and cosmetics.
3. React ethanoic acid with ethanol to produce an ester for fragrances, and study its reaction mechanism.
4. Evaluate the use of the esters and amines in producing fragrances and cosmetics

You have been tasked by your teacher to design feasible chemical processes, explain the underlying organic chemistry, and propose sustainable solutions to help the plant reduce environmental harm while maintaining profitability.

**ITEM 6**

In southwestern Uganda, a small cosmetics start-up is producing herbal skin-care creams using locally sourced plant oils, such as shea butter and sunflower oil. The production process generates significant amounts of waste plant oils and fats. These wastes are currently disposed of into nearby drainage systems, causing blockages and foul smells, which has led to complaints from the community and environmental authorities.

The company has approached a team of A-level chemistry students in your school to:

1. Analyse the composition of these waste materials, their functional groups and physical properties,
2. Explore possible chemical processes to convert them into valuable products and their mechanisms,
3. Design a synthetic route to convert compound C into compound D
4. Propose a sustainable chemical process to convert waste A into a useful, marketable product that reduces environmental harm.

Analysis of a sample of the waste oil revealed the presence of a mixture of the following compounds:

| Compound | Structure (condensed) |
|---|---|
| A | $CH_3(CH_2)_{14}COOCH_2CH(OH)CH_2OH$ |
| B | $CH_3(CH_2)_7CH{=}CH(CH_2)_7COOH$ |
| C | $CH_3CH_2CH_2OH$ |
| D | $CH_3COCH_3$ |

**Tasks:**

As a chemistry learner, give written presentation that addresses the company's challenges and outlines scientifically sound, sustainable solutions.

# THE PERIODIC TABLE

| 1 | 2 | | | | | | | | | | | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1.0 H 1 | | | | | | | | | | | | | | | | 1.0 H 1 | 4.0 He 2 |
| 6.9 Li 3 | 9.0 Be 4 | | | | | | | | | | | 10.8 B 5 | 12.0 C 6 | 14.0 N 7 | 16.0 O 8 | 19.0 F 9 | 20.2 Ne 10 |
| 23.0 Na 11 | 24.3 Mg 12 | | | | | | | | | | | 27.0 Al 13 | 28.1 Si 14 | 31.0 P 15 | 32.1 S 16 | 35.4 Cl 17 | 40.0 Ar 18 |
| 39.1 K 19 | 40.1 Ca 20 | 45.0 Sc 21 | 47.9 Ti 22 | 50.9 V 23 | 52.0 Cr 24 | 54.9 Mn 25 | 55.8 Fe 26 | 58.9 Co 27 | 58.7 Ni 28 | 63.5 Cu 29 | 65.7 Zn 30 | 69.7 Ga 31 | 72.6 Ge 32 | 74.9 As 33 | 79.0 Se 34 | 79.9 Br 35 | 83.8 Kr 36 |
| 85.5 Rb 37 | 87.6 Sr 38 | 88.9 Y 39 | 91.2 Zr 40 | 92.9 Nb 41 | 95.9 Mo 42 | 98.9 Te 43 | 101 Ru 44 | 103 Rh 45 | 106 Pd 46 | 108 Ag 47 | 112 Cd 48 | 115 In 49 | 119 Sn 50 | 122 Sb 51 | 128 Te 52 | 127 I 53 | 131 Xe 54 |
| 133 Cs 55 | 137 Ba 56 | 139 La 57 | 178 Hf 72 | 181 Ta 73 | 184 W 74 | 186 Re 75 | 190 Os 76 | 192 Ir 77 | 195 Pt 78 | 197 Au 79 | 201 Hg 80 | 204 Ti 81 | 207 Pb 82 | 209 Bi 83 | 209 Po 84 | 210 At 85 | 222 Rn 86 |
| 223 Fr 87 | 226 Ra 88 | 227 Ac 89 | | | | | | | | | | | | | | | |
| | | | 139 La 57 | 140 Ce 58 | 141 Pr 59 | 144 Nd 60 | 147 Pm 61 | 150 Sm 62 | 152 Eu 63 | 157 Gd 64 | 159 Tb 65 | 162 Dy 66 | 165 Ho 67 | 167 Er 68 | 169 Tm 68 | 173 Yb 70 | 175 Lu 71 |
| | | | 227 Ac 89 | 232 Th 90 | 231 Pa 91 | 238 U 92 | 237 Np 93 | 244 Pu 94 | 243 Am 95 | 247 Cm 96 | 247 Bk 97 | 251 Cf 98 | 254 Es 99 | 257 Fm 100 | 256 Md 101 | 254 No 102 | 260 Lw 103 |

**END**